MD. MEHERAZ HOSSAIN CHAYAN

Computer Science Researcher | HCI · Human-Centered AI · Mixed-Methods Research

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RESEARCH PROFILE

BRAC University Computer Science graduate with research experience spanning human-computer interaction, human-centered AI, privacy and digital trust. Experienced in survey and interview research, data analysis, research prototyping and usability evaluation. Recent projects examine bounded AI agents for university students, AI-supported mental-health crisis response and sensitive online reporting systems.

EDUCATION

BSc in Computer Science, BRAC University | *2022–2026* | *Completed* | *CGPA 3.46/4.00*
Thesis: Security and Usability of a Failure-Resilient News Provenance System in a Fragile Information Ecosystem: Architecture and Controlled User Evaluation.

Relevant courses: Human-Computer Interaction (CSE472, A); Artificial Intelligence (CSE422, A); Machine Learning (CSE427, A); Thesis (CSE400, A); Software Engineering (CSE470, A−); Computer Architecture (CSE340, A); Introduction to Football Data and Analytics (CSE490D, A−).

RESEARCH EXPERIENCE

Research Associate & Developer, University of Central Florida (Remote) | *Jan–Jul 2026*
Supervisor: Dr. Mohammad Newaz Sharif

- Developed RES-Q, a machine-learning-based mental-health monitoring and crisis-response research system for emergency professionals and affected communities.
- Contributed to user research, iterative prototyping, system evaluation and multidisciplinary research documentation; project selected as a Top 8 finalist at the Johns Hopkins Healthcare Design Competition 2026.

Undergraduate Researcher, University of Canberra (Remote) | *Sep 2024–May 2025*
Supervisor: Dr. Md. Mizanur Rahman

- Developed and evaluated ML approaches for smart-home and IoT cyberattack detection, including dataset preparation, comparative model evaluation and explainability-focused analysis.
- Analyzed IoT cyberattack datasets and interpreted model features to explain detection outcomes.
- Prepared comparative research findings and technical reports documenting model performance and evaluation.

ADDITIONAL TECHNICAL EXPERIENCE

Avionics & Control Systems R&D Engineer, AERD | *Oct 2023–Aug 2024*

- Developed SkyRecon networking-drone concepts involving mesh routing, relay coordination and link monitoring; also contributed to electronic valve control, flight-control algorithms and integration testing.

Machine Learning Intern, Interactive Cares | *Jan–Mar 2025*

- Preprocessed datasets, trained and evaluated supervised learning models, and documented results.

Web Development Intern, Interactive Cares | *Feb–Apr 2023*

- Built responsive React/JavaScript interfaces and supported full-stack development.

RESEARCH METHODS & TECHNICAL SKILLS

Methods: Survey and interview design, usability studies, qualitative/statistical analysis, experimental design, dataset preparation, model comparison, academic writing.

AI & Data: Python, SQL, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, NLP, model evaluation and explainable AI.

HCI & Development: Human–AI interaction, responsible AI, privacy and trust, React, APIs, Git, LLM/RAG and agentic workflows.

SELECTED PUBLICATIONS & RESEARCH OUTPUTS

Published | *ACM NSysS 2024*
Md Mizanur Rahman, Md Meheraz Hossain Chayan, Khaleda Mehrin, Afrin Sultana, and Mosaab M. Hamed. “Explainable Deep Learning for Cyber Attack Detection in Electric Vehicle Charging Stations.” Proceedings of NSysS ’24, ACM, pp. 1–7. DOI: 10.1145/3704522.3704534.

Accepted |

- “Data for Protection, Data as Risk: Governing Humanitarian Information in Bangladesh’s Rohingya Response.” 2027 ANZ and Pacific Conference.

- “Incremental Cancer-Class Expansion from HAM10000 to MILK10k with External Validation on DDI/DermaBench.” Computers in Biology and Medicine (Elsevier).

Submitted |

- “An AI That Knows Its Limits: Designing Bounded Agents for University Students.” ACM CHI 2027.
- “From Claims to Evidence: Designing and Evaluating a Multimodal News Verification System.” ACM CHI 2027.
- “Reporting Without Fear: Designing for Trust and Safety in Digital Bribery Reporting.” ACM CHI 2027.
- “ReliableVAEP: Bootstrap Uncertainty and Probabilistic Player Rankings for Football Action Valuation.” MIT Sloan Sports Analytics Conference 2027.

Manuscripts in Preparation |
Agentic AI for Cyberattack Detection in IoT Devices (target: Computers & Security); Dataset on User-Reported Bribery Cases in Bangladesh for Machine Learning/Deep Learning Studies (target: Scientific Data); A Standard for Consistently Collected Data on Real and Fake News Claims (target: Scientific Data).

SELECTED RESEARCH PROJECTS

My University Portal | *HCI · Bounded AI · Student Services*
Mixed-methods study involving 209 survey participants and 30 interviews across private universities; designed a task-scoped AI prototype emphasizing approval controls, explanations and user privacy.

RES-Q | *Mental-Health AI · Crisis Response*
ML-based mental-health monitoring and crisis-response prototype; contributed to user research, design, implementation and iterative evaluation.

NewsProof | *Multimodal Verification · Usability*
Built a system for inspecting news claims and evidence across URLs, text, images and audio, with a human-centered evaluation of verification and user decision-making.

Ghush | *Privacy · Human-Centered Reporting*
Designed an anonymous bribery-reporting platform with AI-assisted filtering, human review and safety-oriented feedback mechanisms.

CancerIL / HomeGuard XAI | *Continual Learning · Explainable Cybersecurity*
CancerIL studies incremental skin-lesion classification and cross-dataset evaluation; HomeGuard XAI compares interpretable cyberattack-detection methods for IoT environments.

RESEARCH-BACKED TECHNICAL PROJECTS

- SkyRecon: Networking-drone research prototype with multi-hop routing, relay coordination and link monitoring.
- HealthLinkedAI: AI-enabled healthcare prototype combining disease detection and patient–doctor support.
- MediVerify: Medical-data integrity system for detecting manipulated records and explaining verification risks.
- Social Square: Digital platform supporting collaborative brand promotions and marketing opportunities.
- MedBot Assist: Assistive mobile robotics project designed to support people with disabilities.
- Care Cloud: MERN-based healthcare platform for patient, doctor and medical-record workflows.
- Mini F1: Interactive racing simulation featuring real-time controls and OpenGL graphics.
- BlockAware: Blockchain-based digital-currency awareness project; national contest third-place entry.

SELECTED ACHIEVEMENTS

- Top 8 Finalist, Johns Hopkins Healthcare Design Competition (2026), RES-Q; Johns Hopkins CBID.
- 3rd Place, National Undergraduate Startup Challenge (2026), Social Square; Eastern Bank PLC.
- 3rd Place, National Blockchain Hackathon and Video Awareness Contest (2026), BlockAware; BSEC.
- National Champion, International Science and Invention Fair Tech & Robotics Competition (2025), MediVerify; IYSA.
- Winner, UIHP National Innovation Challenge (2025), HealthLinkedAI; UIHP / ICT Division / Bangladesh Hi-Tech Park Authority.
- Champion, National Research Essay Competition on World Environment Day (2023); Dhaka School of Economics.

CERTIFICATIONS & LANGUAGES

Machine Learning Specialization (DeepLearning.AI); Google Data Analytics Professional Certificate; CCNA; Microsoft AI Product Manager Professional Certificate. Bengali (native); English (IELTS Academic 7.0).

ACADEMIC REFERENCES

Dr. Farida Chowdhury (BRAC University) | Dr. Muhammad Iqbal Hossain (BRAC University) | Dr. Mohammad Newaz Sharif (University of Central Florida) | Dr. Md. Mizanur Rahman (University of Canberra).